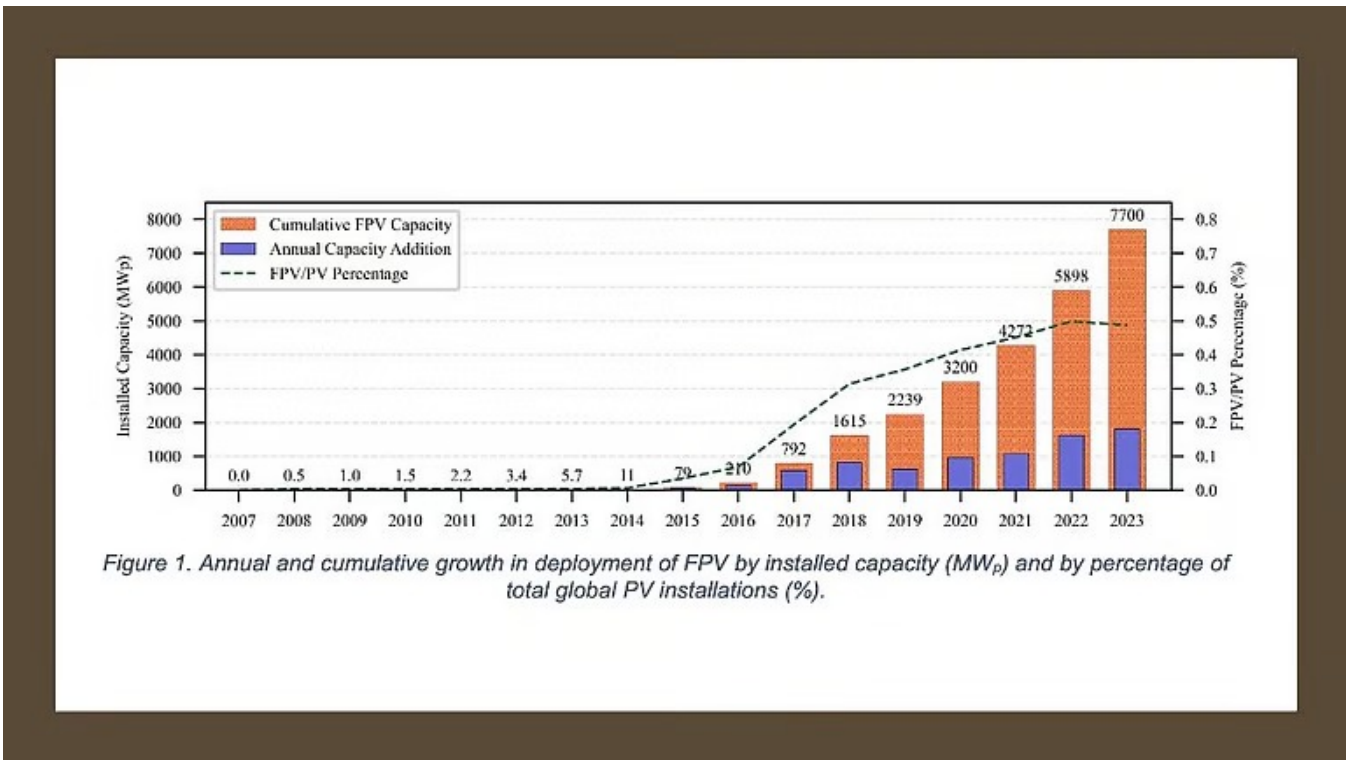




Technology

IEA PVPS: Global Floating PV Capacity Hit 7.7 GW In 2023

New report attempts to address knowledge gaps for the sustainable development of the floating PV application



Asia, led by China, leads the global installed floating PV capacity which reached 7.7 GW at the end of 2023, according to the IEA PVPS report. (Photo Credit: IEA PVPS)

 Anu Bhambhani

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Key Takeaways

- *China leads the world's floating PV capacity, which expanded to 7.7 GW by the end of 2023*
- *IEA PVPS addresses the environmental impact and the lack of quantitative data for this application that hinders its growth*
- *It calls for increased R&D in the space to improve materials and reduce expenses by investing in AI tools*

The global floating PV (FPV) installed capacity expanded to 7.7 GW at the end of 2023, with almost 90% located in Asia, out of which China alone commands a leading position with a 50% share, and interest in such projects continues to increase.

However, as the International Energy Agency's Photovoltaic Power Systems Programme (IEA-PVPS) points out, there are still several uncertainties about the environmental impacts and system reliability of FPV technology. This is due to the absence of regulatory frameworks and limited long-term data.

While there are guidelines published by the World Bank, DNV, SolarPower Europe (SPE) and national level standards in South Korea, China and Singapore, the IEA PVPS says there is a lack of quantitative data for yield modeling and reliability in this space. This is what it attempts to address in its new Task 13 report titled *Floating Photovoltaic Power Plants: A Review of Energy Yield, Reliability, and Maintenance* while providing practical guidance for stakeholders.

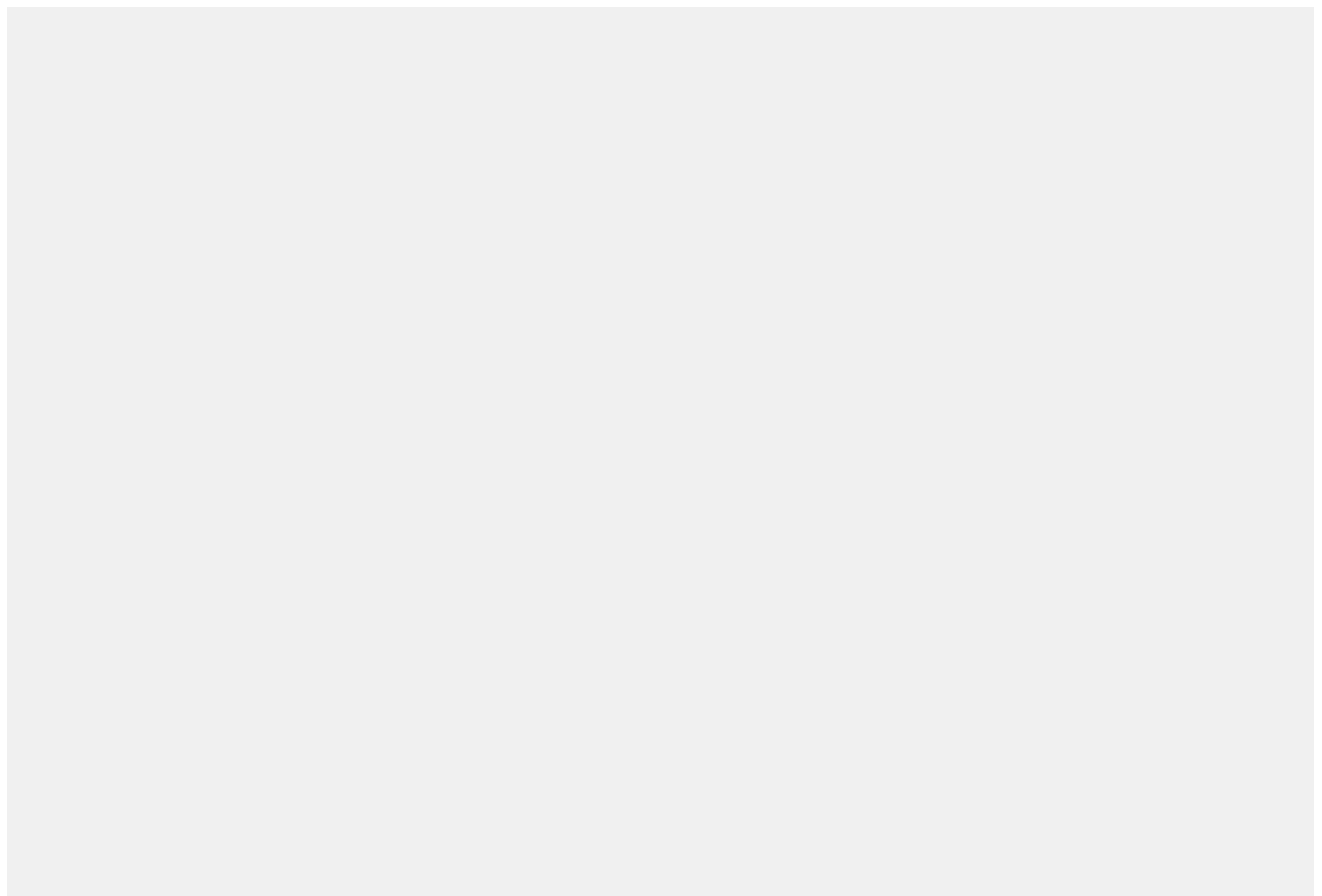
“Rapid innovation in the field often prioritizes confidentiality, limiting data sharing crucial for industry growth. This report aims to support FPV

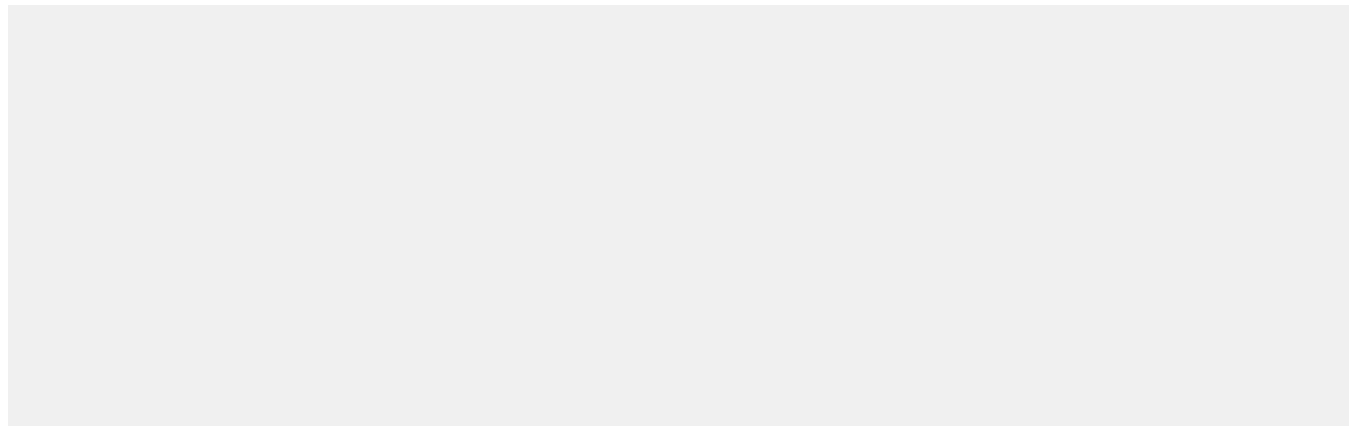
development by building a knowledge base on energy yield, reliability, and O&M — areas where FPV diverges from GPV,” reads the report.

China, as the world’s largest FPV market, is followed by Taiwan, India, Israel, Japan and South Korea. However, the report writers believe that FPV holds potential to support the EU’s climate neutrality goals, with the Netherlands and France currently hosting the 7th and 10th largest FPV capacities.

Currently, most of the FPV systems are installed on sheltered inland waters, including quarry lakes, irrigation ponds and reservoirs. Existing meteorological databases often exclude sea and coastal areas that limit FPV’s energy yield assessment (EYA). The current EYA models do not offer reliable data to study critical parameters like module temperature, wave-induced losses, soiling losses and performance loss rates.

This is key to determining the levelized cost of electricity and project profitability, according to the report writers. Hence, they recommend improving EYA accuracy for FPV by closing gaps in meteorological data to understand how variables like irradiance, wind, and temperature affect the prediction for such systems.





IEA PVPS counts Sungrow as leading the floating PV technology suppliers globally with a 27% share. (Photo Credit: IEA PVPS)

One of the recommendations is to improve the modeling tools, such as PVsyst and pplib, to get more accurate yield estimates to improve thermal performance, since the latter depends on the system design.

Another missing link is that there is currently no complete model for wave-induced losses, for which the IEA PVPS report writers believe more field data must be collected to improve accuracy in yield modeling.

While FPV systems are not generally considered prone to soiling, analysts believe that such systems may experience unique soiling challenges, including bird droppings and other debris from surrounding ecosystems, which may impact the system's performance.

As more onshore and offshore FPV projects are planned, their operation and maintenance (O&M) framework becomes important to work upon. The report recommends the following R&D possibilities to make the investment work:

- **Monitoring and Remote Sensing:** Remote FPV sites face challenges with data transmission and costs, but advanced tools like drones, satellites, IoT, and cloud storage can improve monitoring and cut down operational expenses.
- **Expert Dependence:** Maintaining FPV systems is costly due to reliance on specialists, but AI, drones, and automation can cut the need for human intervention.
- **Extreme Weather and Degradation:** Harsh marine conditions speed up FPV wear and tear, making advanced materials, better designs, and emergency planning essential for long-term durability.

- **Environmental Impact and Regulations:** FPV's impact on aquatic ecosystems calls for eco-friendly designs and regulations to ensure sustainability.

“By addressing these research priorities, the industry can move towards a more mature and sustainable deployment of FPV, ultimately paving the way for its widespread adoption,” according to the report.

In terms of technology providers for FPV floating system manufacturers, China’s Sungrow leads the space with a 27% global market share in terms of installed capacity, followed by Northman Energy Technology with 14.9%, and France’s Ciel & Terre with a 12.7% share. Since various float types are in the market, IEA PVPS recommends slotting them in the following 3 basic categories to be able to measure their expected performance:

- Pure pontoon-based floats, made with high-density polyethylene (HDPE)
- Pontoon floats+ metal/FRP, comprising metal or fiber-reinforced plastic (FRP) with floats or pipes, for instance ZIM Float of Europe, and
- Non-pontoon-based floats such as platforms, ferrocement structures, and membrane technology. The latter is patented by Ocean Sun.

The complete IEA PVPS report is available for free download on its [website](#).

TaiyangNews published its 1st *Market Survey Floating PV 2024* featuring the commercial products available in this space. The survey can be downloaded for free [here](#).

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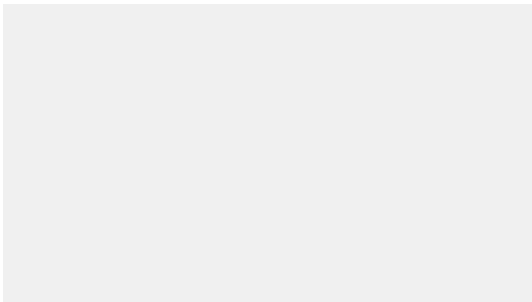


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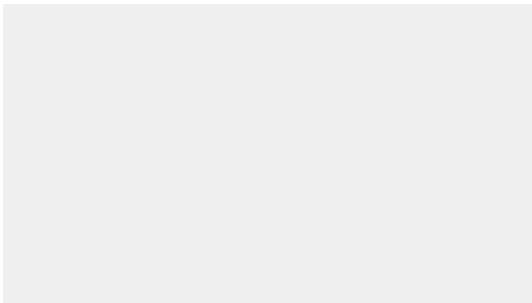
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